

Learning Objectives

- Relate the amount of heat released or absorbed by a reaction using a balanced chemical reaction and the reaction stoichiometry.
- Use ΔH_f to determine the ΔH_{rxn} for a reaction.
- Calculate the ΔE_{rxn} with heat and work for a chemical reaction.
- Use bond energies and Lewis structure for a chemical reaction to calculate an approximate ΔH for a chemical reaction.
- Determine if the reaction is exothermic or endothermic from the chemical PE and the bond strength of the reactants and products.

Enthalpy (H): Heat of Reaction

Enthalpy: The sum of the internal Energy and the P-V work of a system under constant pressure.

$$H = E + PV$$

- Change of enthalpy under constant pressure is:

$$\Delta H = \Delta E + P\Delta V$$

$$\Delta E = q_p + w$$

$$\Delta H = q_p + (-P\Delta V) + P\Delta V$$

$$w = -P\Delta V$$

$$\Delta H = q_p$$

- Enthalpy is a state function. $\Delta H = H_{\text{final}} - H_{\text{initial}}$

Your Turn! $\Delta H = \Delta E + P\Delta V$

$$\Delta H = q_p$$

When 1 mole of butane (C_4H_{10}) from a lighter burns at constant pressure, the reaction produces 2658 kJ of heat and does 3 kJ of PV work. What are the values of ΔH and ΔE for the combustion of 1 mole of butane?

☒ A. $\Delta H = -2658 \text{ kJ}$; $\Delta E = -2661 \text{ kJ}$

B. $\Delta H = -2658 \text{ kJ}$; $\Delta E = -2655 \text{ kJ}$

☒ C. $\Delta H = +2658 \text{ kJ}$; $\Delta E = +2661 \text{ kJ}$

☒ D. $\Delta H = +2658 \text{ kJ}$; $\Delta E = +2655 \text{ kJ}$

$$\begin{aligned}\Delta E &= \Delta H - P\Delta V \\ &= \Delta H + w \\ &= -2658 \text{ kJ} + (-3 \text{ kJ})\end{aligned}$$



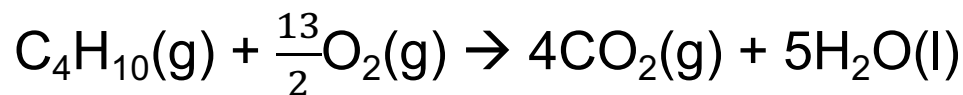
Enthalpy of Reaction (ΔH_{rxn})

Difference in enthalpy between products and reactants according to the balanced chemical equation.

- $\Delta H_{\text{rxn}} = H_{\text{final}} - H_{\text{initial}}$
 $= \sum H_{\text{(product)}} - \sum H_{\text{(reactants)}}$
- Most of reactions occur under constant pressure so $\Delta H = q_p$
Endothermic reaction: $q_p \geq 0$ (+) then $\Delta H \geq 0$ (+)

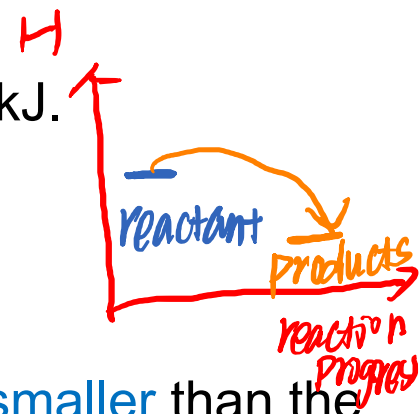
Exothermic reaction: $q_p \leq 0$ (-) then $\Delta H \leq 0$ (-)

ex. ΔH_{rxn} of 1 mole of butane (C_4H_{10}) combusts in air is - 2658 kJ.



Combustion of butane is Endothermic / Exothermic reaction.

In this reaction, the sum of the reactants' enthalpy are larger / smaller than the products' enthalpy.

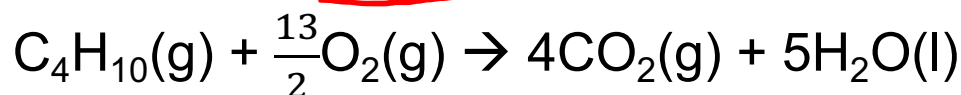


Enthalpy of Reaction (ΔH_{rxn})

ΔH_{rxn} is an extensive property (addable).

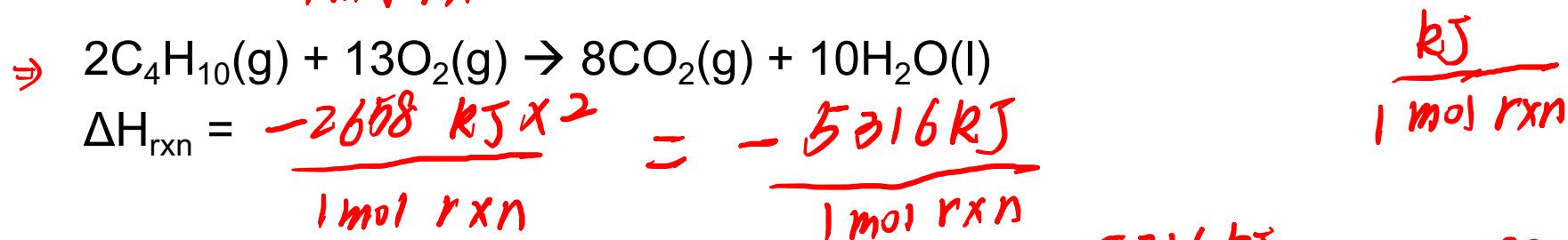
- ΔH_{rxn} depends on the coefficients in the chemical equation.

Ex. ΔH_{rxn} of 1 mole of butane (C_4H_{10}) combusts in air is - 2658 kJ.



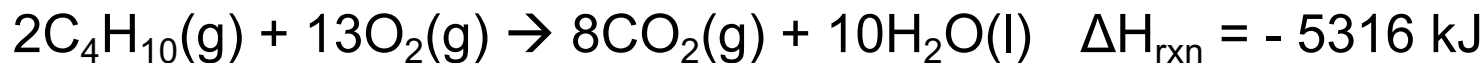
$$\Delta H_{\text{rxn}} = \frac{-2658 \text{ kJ}}{1 \text{ mol rxn}}$$

$$\Delta H = \text{kJ}$$



- ΔH_{rxn} follows stoichiometry.

$$-5316 \text{ kJ} = 8 \text{ mol CO}_2$$



$$-5316 \text{ kJ} = \underline{2} \text{ mol C}_4\text{H}_{10}(\text{g}) = \underline{13} \text{ mol O}_2(\text{g}) = \underline{8} \text{ mol CO}_2(\text{g}) = \underline{10} \text{ mol H}_2\text{O}(\text{l})$$

If 400. kJ heat is produced from the reaction, how much, in mol, of CO_2 is produced?

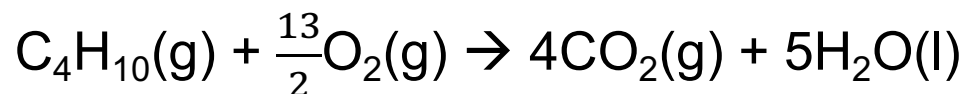
$$-400. \text{ kJ} \times \frac{8 \text{ mol CO}_2}{-5316 \text{ kJ}} = 0.602 \text{ mol of CO}_2$$

Enthalpy of Reaction (ΔH_{rxn})

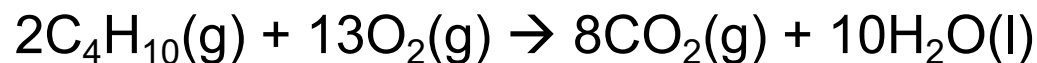
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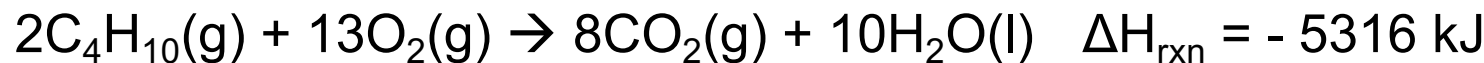


$$\Delta H_{\text{rxn}} = -2658 \text{ kJ}$$



$$\Delta H_{\text{rxn}} = -2658 \text{ kJ} \times 2 = -5316 \text{ kJ}$$

- ΔH_{rxn} follows stoichiometry.



$$-5316 \text{ kJ} = \underline{2} \text{ mol C}_4\text{H}_{10}(\text{g}) = \underline{13} \text{ mol O}_2(\text{g}) = \underline{8} \text{ mol CO}_2(\text{g}) = \underline{10} \text{ mol H}_2\text{O}(\text{l})$$

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$$\underline{(-400. \text{ kJ})} \times \frac{8 \text{ mol CO}_2}{-5316 \text{ kJ}} = 0.602 \text{ mol CO}_2$$

Announcement 2/11

Assignments that are due soon

- Pre-Class Assignment Week 2-F

Chem RePS Session

Wed 2:00 - 3:30 PM at KC 230

1st Chem RePS or SI worksheets for Unit 1 will be closed on 2/18 at 11:59pm

1st bonus

Wed

Office Hour

- Dr. Wang's: Thu 10-11 AM on Zoom

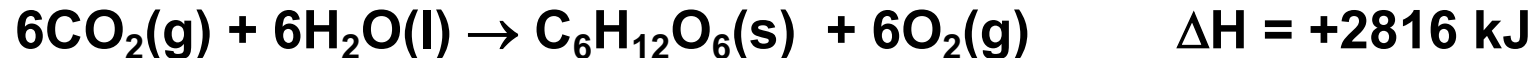
SI Session

Saige Douglass – Wed 5:30 – 7:30 PM at Hashinger 412

Paige Kwan - Thu 4 - 6 PM at Hashinger 412

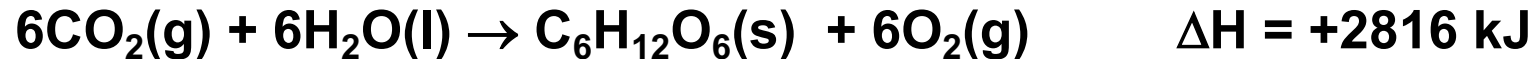
Your Turn!

With 100. kJ of energy are provided, how much CO₂ (molar mass = 44.01 g/mol) , in grams, can be converted to glucose according to the following reaction?



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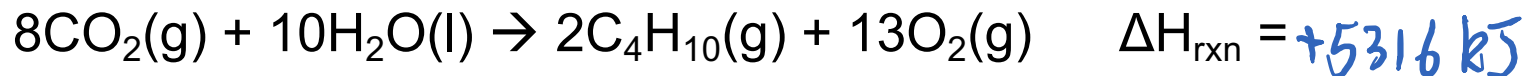
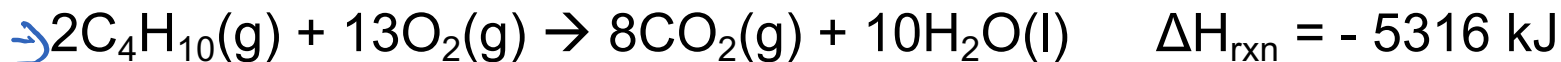


$$\begin{aligned} & \text{[+100. kJ]} \times \frac{6 \text{ mol CO}_2}{+2816 \text{ kJ}} \times \frac{44.01 \text{ g CO}_2}{1 \text{ mol CO}_2} = 9.38 \text{ g CO}_2 \\ & \text{= } \end{aligned}$$

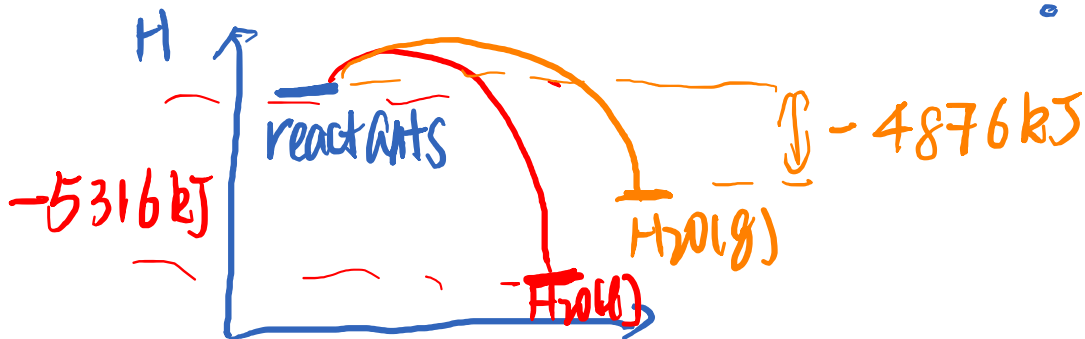
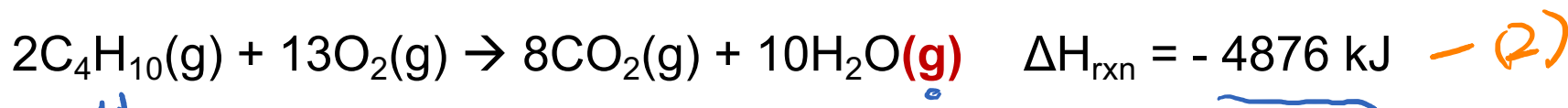
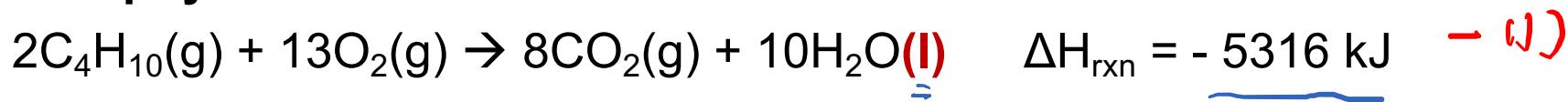
1 mol reaction

Enthalpy of Reaction (ΔH_{rxn})

- If the direction of the chemical equation is reversed, then ΔH_{rxn} reverses sign.



- The **physical states** of the chemicals matters.



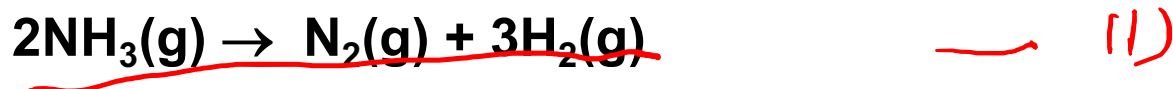
Which physical state of water is more stable?

$\text{H}_2\text{O}(\text{l})$



Your Turn!

When 5.000 g of gaseous H_2 (molar mass = 2.016 g/mol) reacts with excess amount of gaseous N_2 to form gaseous NH_3 under constant pressure and produce 76.37 kJ of heat. What is the ΔH_{rxn} , in kJ, for 1 mol of the following reaction?



Method 1: $-76.37 \text{ kJ} = 5.000 \text{ g H}_2$

$$\Delta H_{\text{rxn}2} = \frac{-76.37 \text{ kJ}}{5.000 \text{ g H}_2} \times \frac{2.016 \text{ g}}{1 \text{ mol H}_2} \times \frac{3 \text{ mol H}_2}{1 \text{ mol rxn}} = \frac{-92.38 \text{ kJ}}{1 \text{ mol rxn}}$$

Method 2: $\Delta H_{\text{rxn}2} = x$

$$\underbrace{5.00 \text{ g}}_{m(\text{H}_2)} \times \frac{1 \cancel{\text{ mol H}_2}}{2.016 \cancel{\text{ g}}} \times \frac{1 \cancel{\text{ mol rxn}}}{3 \cancel{\text{ mol H}_2}} \times \frac{\Delta H_{\text{rxn}2}}{1 \cancel{\text{ mol rxn}}} = -76.37 \text{ kJ}$$

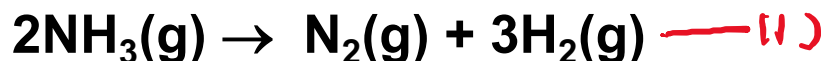
$$= 9 (5.00 \text{ g H}_2)$$

$$\Delta H_{\text{rxn}(2)} = -92.38 \text{ kJ}$$

$$\Delta H_{\text{rxn}(1)} = -\Delta H_{\text{rxn}(2)} = +92.38 \text{ kJ}$$

Your Turn!

When 5.000 g of gaseous H_2 (molar mass = 2.016 g/mol) reacts with excess amount of gaseous N_2 to form gaseous NH_3 under constant pressure and produce 76.37 kJ of heat. What is the ΔH_{rxn} , in kJ, for 1 mol of the following reaction? q_p



$$\Delta H_{\text{rxn(2)}} = q_p = \frac{-76.37 \text{ kJ}}{5.000 \text{ g H}_2} \times \frac{2.016 \text{ g H}_2}{1 \text{ mol H}_2} \times 3 \text{ mol H}_2$$
$$= -92.38 \text{ kJ}$$

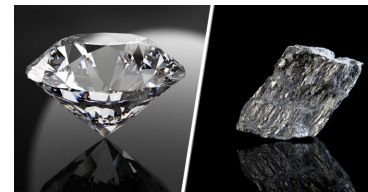
$$\Delta H_{\text{rxn(1)}} = -\Delta H_{\text{rxn(2)}} = -(-92.38 \text{ kJ}) = +92.38 \text{ kJ}$$

Standard Reaction Enthalpy $\Delta H_{\text{rxn}}^\circ$

- Standard states ($^\circ$):

$$T = \underline{25}^\circ\text{C}$$

$$\text{For gas: } P = \underline{10^5} \text{ Pa}$$



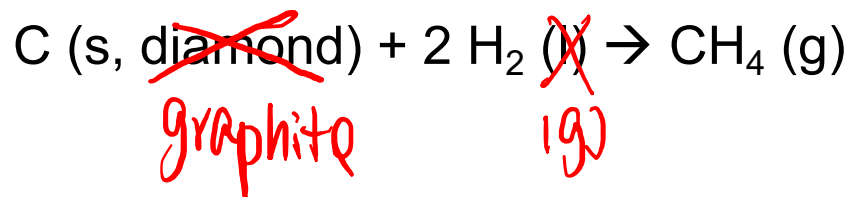
For liquid and solid: most stable form at 25°C (ex. Carbon: graphite)

$$\text{For solution: } M = \underline{1} \text{ M}$$

- Standard enthalpy of reaction ($\Delta H_{\text{rxn}}^\circ$):

Enthalpy change of the reaction that takes place under standard conditions

Modify the following equation so that the enthalpy of the reaction is $\Delta H_{\text{rxn}}^\circ$:



Standard Enthalpy of Formation ΔH_f°

- Standard enthalpy of formation (ΔH_f°):

Enthalpy change for the reaction forming 1 mole of a pure compound from its constituent elements in their standard states.

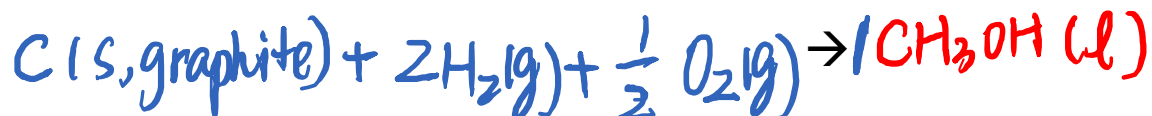
- Unit for ΔH_f° : $\frac{\text{kJ}}{1 \text{ mol of product}}$

TABLE 9.4 Standard Enthalpies (or Heats) of Formation, ΔH_f° , at 298 K

Formula	ΔH_f° (kJ/mol)	Formula	ΔH_f° (kJ/mol)	Formula	ΔH_f° (kJ/mol)
Bromine		$\text{C}_3\text{H}_8\text{O}(l, \text{isopropanol})$	-318.1	Oxygen	
$\text{Br}(g)$	111.9	$\text{C}_6\text{H}_6(l)$	49.1	$\text{O}_2(g)$	0
$\text{Br}_2(l)$	0	$\text{C}_6\text{H}_{12}\text{O}_6(s, \text{glucose})$	-1273.3	$\text{O}_3(g)$	142.7
$\text{HBr}(g)$	-36.3	$\text{C}_{12}\text{H}_{22}\text{O}_{11}(s, \text{sucrose})$	-2226.1	$\text{H}_2\text{O}(g)$	-241.8
Calcium		Chlorine		$\text{H}_2\text{O}(l)$	-285.8
$\text{Ca}(s)$	0	$\text{Cl}(g)$	121.3	Silver	
$\text{CaO}(s)$	-634.9	$\text{Cl}_2(g)$	0	$\text{Ag}(s)$	0
$\text{CaCO}_3(s)$	-1207.6	$\text{HCl}(g)$	-92.3	$\text{AgCl}(s)$	-127.0
Carbon		Fluorine		Sodium	
$\text{C}(s, \text{graphite})$	0	$\text{F}(g)$	79.38	$\text{Na}(s)$	0
$\text{C}(s, \text{diamond})$	1.88	$\text{F}_2(g)$	0	$\text{Na}(g)$	107.5
$\text{CO}(g)$	-110.5	$\text{HF}(g)$	-273.3	$\text{NaCl}(s)$	-411.2
$\text{CO}_2(g)$	-393.5	Hydrogen		$\text{Na}_2\text{CO}_3(s)$	-1130.7
$\text{CH}_4(g)$	-74.6	$\text{H}(g)$	218.0	$\text{NaHCO}_3(s)$	-950.8
$\text{CH}_3\text{OH}(l)$	-238.6	$\text{H}_2(g)$	0	Sulfur	

- For any pure element in its standard state: $\Delta H_f^\circ = 0$ kJ/mol
- A more negative ΔH_f° standards for a more stable substance

Write balanced formation equation for $\text{CH}_3\text{OH}(l)$:



Reactants: pure element in its standard state

Product: one mol of the compound

$$\Delta H_f^\circ = \frac{-238.6 \text{ kJ}}{1 \text{ mol CH}_3\text{OH}(l)}$$

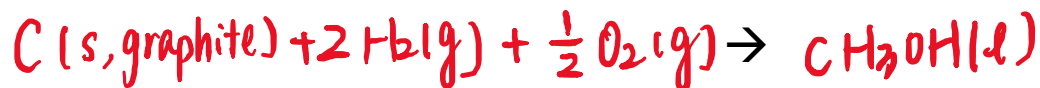
Standard Enthalpy of Formation ΔH_f°

- Standard enthalpy of formation (ΔH_f°):
Enthalpy change for the reaction forming 1 mole of a pure compound from its constituent elements in their standard states.
- Unit for ΔH_f° : $\frac{\text{kJ}}{\text{mol}}$ $\frac{\Delta H^\circ}{1 \text{ mol of compound}}$
- For any pure element in its standard state: $\Delta H_f^\circ = \underline{0}$ kJ/mol

TABLE 9.4 Standard Enthalpies (or Heats) of Formation, ΔH_f° , at 298 K

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Bromine		$\text{C}_3\text{H}_8\text{O}(l, \text{isopropanol})$	-318.1	Oxygen	
$\text{Br}(g)$	111.9	$\text{C}_6\text{H}_6(l)$	49.1	$\text{O}_2(g)$	0
$\text{Br}_2(l)$	0	$\text{C}_6\text{H}_{12}\text{O}_6(s, \text{glucose})$	-1273.3	$\text{O}_3(g)$	142.7
$\text{HBr}(g)$	-36.3	$\text{C}_{12}\text{H}_{22}\text{O}_{11}(s, \text{sucrose})$	-2226.1	$\text{H}_2\text{O}(g)$	-241.8
Calcium		Chlorine		$\text{H}_2\text{O}(l)$	-285.8
$\text{Ca}(s)$	0	$\text{Cl}(g)$	121.3	Silver	
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$\text{CaCO}_3(s)$	-1207.6	$\text{HCl}(g)$	-92.3	$\text{AgCl}(s)$	-127.0
Carbon		Fluorine		Sodium	
$\text{C}(s, \text{graphite})$	0	$\text{F}(g)$	79.38	$\text{Na}(s)$	0
$\text{C}(s, \text{diamond})$	1.88	$\text{F}_2(g)$	0	$\text{Na}(g)$	107.5
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$\text{CH}_3\text{OH}(l)$	-238.6	$\text{H}_2(g)$	0	Sulfur	

Write balanced formation equation for $\text{CH}_3\text{OH}(l)$:



$$\Delta H = \sum H_{\text{product}} - \sum H_{\text{reactant}}$$

$$\Delta H_f^\circ = -238.6 \frac{\text{kJ}}{\text{mol}}$$

Reactants: pure element in its standard state Product: one mol of the compound 21

Your Turn!

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Calcium		C ₁₂ H ₂₂ O ₁₁ (s, sucrose)	-2226.1	H ₂ O(l)	-285.8
		Chlorine		Silver	
Ca(s)	0	Cl(g)	121.3	Ag(s)	0
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Carbon		Fluorine			
C(s, graphite)	0	F(g)	79.38	Na(s)	0
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CH ₃ OH(l)	-238.6	H ₂ (g)	0	Sulfur	
C ₂ H ₂ (g)	227.4	Nitrogen		S ₈ (s, rhombic)	0
C ₂ H ₄ (g)	52.4	N ₂ (g)	0	S ₈ (s, monoclinic)	0.3
C ₂ H ₆ (g)	-84.68	NH ₃ (g)	-45.9	SO ₂ (g)	-296.8
C ₂ H ₅ OH(l)	-277.6	NH ₄ NO ₃ (s)	-365.6	SO ₃ (g)	-395.7
C ₃ H ₈ (g)	-103.85	NO(g)	91.3	H ₂ SO ₄ (l)	-814.0
C ₃ H ₆ O(l, acetone)	-248.4	N ₂ O(g)	81.6		

Write balanced formation equation for NaHCO₃ (s)

→

$\Delta H_f^\circ =$

Reactant: pure element in its standard state

Product: the compound

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$\text{CH}_3\text{OH}(l)$	-238.6	$\text{H}_2(g)$	0	Sulfur	
$\text{C}_2\text{H}_2(g)$	227.4	Nitrogen		$\text{S}_8(s, \text{rhombic})$	0
$\text{C}_2\text{H}_4(g)$	52.4	$\text{N}_2(g)$	0	$\text{S}_8(s, \text{monoclinic})$	0.3
$\text{C}_2\text{H}_6(g)$	-84.68	$\text{NH}_3(g)$	-45.9	$\text{SO}_2(g)$	-296.8
$\text{C}_2\text{H}_5\text{OH}(l)$	-277.6	$\text{NH}_4\text{NO}_3(s)$	-365.6	$\text{SO}_3(g)$	-395.7
$\text{C}_3\text{H}_8(g)$	-103.85	$\text{NO}(g)$	91.3	$\text{H}_2\text{SO}_4(l)$	-814.0
$\text{C}_3\text{H}_6\text{O}(l, \text{acetone})$	-248.4	$\text{N}_2\text{O}(g)$	81.6		

Write balanced formation equation for $\text{NaHCO}_3(s)$



$$\Delta H_f^\circ = -950.8 \frac{\text{kJ}}{\text{mol}}$$

Reactant: pure element in its standard state

Product: the compound

Announcement 2/13

Assignments that are due soon

- Pre-Class Assignment Week 3-M

Office Hour

- Dr. Wang's: Fri 1-2 PM at KC 382



M.S. Food Science Integrated Program Open House

February 20th, 2026 | 12-2 PM
The Keck Center - McCardle Steps

Join us for an open house to learn
more about Chapman's Integrated
Undergraduate / MS in Food Science
Program!

Discover how you can begin master's
coursework during your senior year
and earn your MS in Food Science just
one year after completing your
bachelor's degree.

Free pizza, lab tours, coffee, and ice
cream!

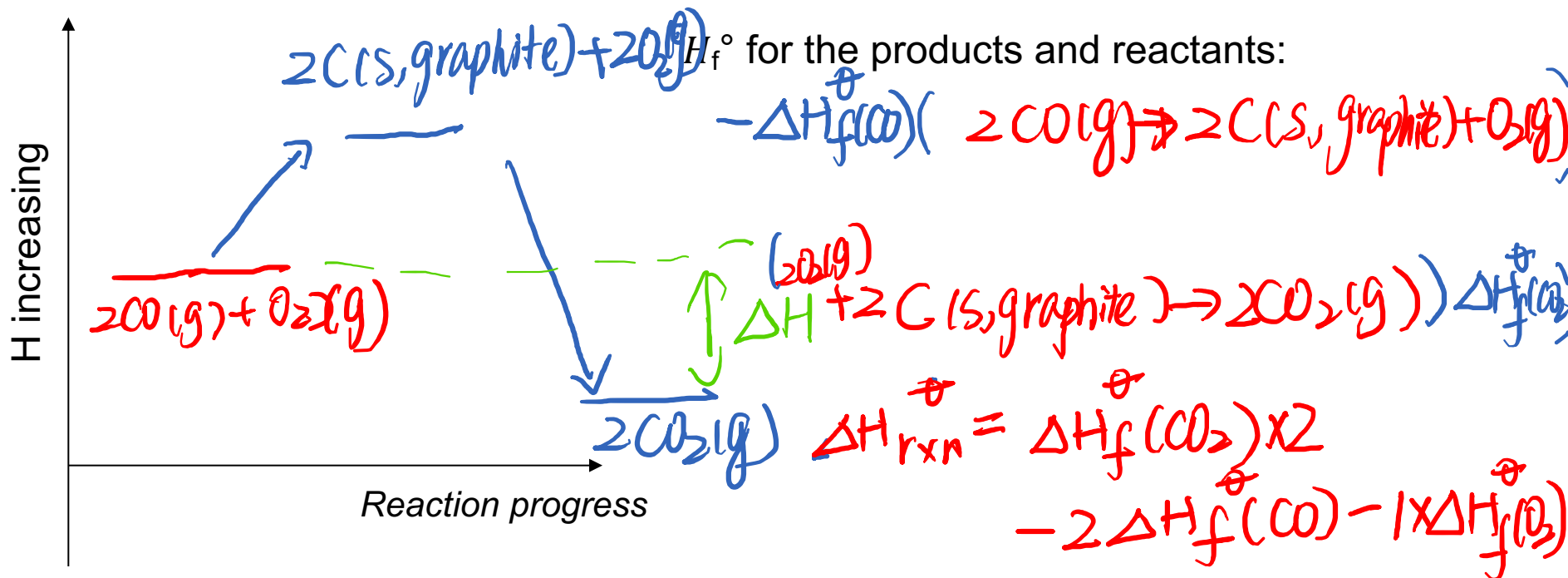
Scan the QR code to RSVP for the
information session and secure your
spot!



Hess's Law: Determine $\Delta H_{\text{rxn}}^\circ$ Indirectly

- Enthalpy (H) is a state function.
- When reactants are converted to products, ΔH is the same whether the reaction takes place in one step or in a series of steps.

Calculate $\Delta H_{\text{rxn}}^\circ$ for reaction: $2\text{CO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{CO}_2\text{(g)}$



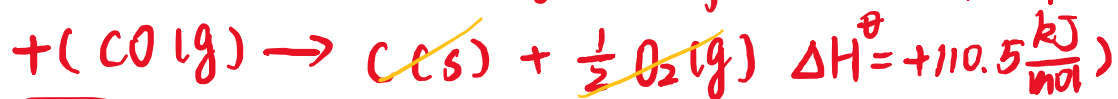
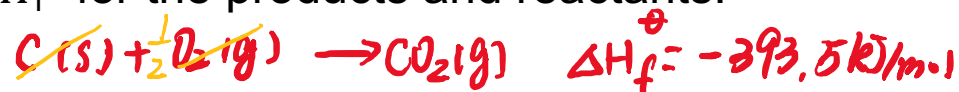
• $\Delta H_{\text{rxn}}^\circ = \sum n \Delta H_{f,\text{products}}^\circ - \sum m \Delta H_{f,\text{reactants}}^\circ$

Hess's Law: Determine $\Delta H_{\text{rxn}}^\circ$ Indirectly

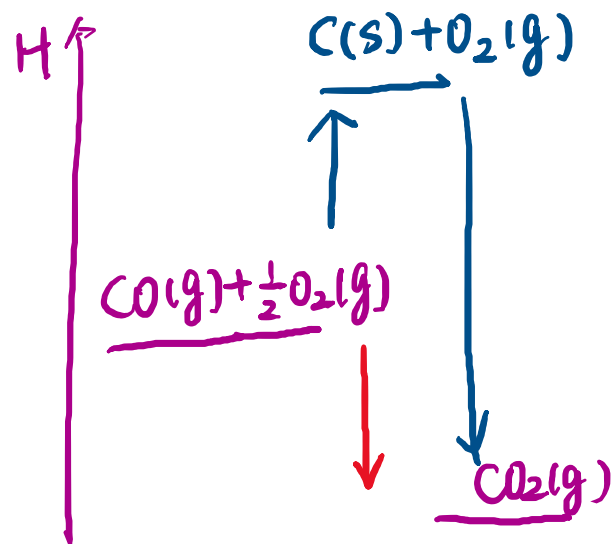
- Enthalpy (H) is a state function.
- When reactants are converted to products, ΔH is the same whether the reaction takes place in one step or in a series of steps.

Calculate $\Delta H_{\text{rxn}}^\circ$ for reaction: $2\text{CO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{CO}_2\text{(g)}$

H_f° for the products and reactants:



$$\Delta H_{\text{rxn}}^\circ = -566 \text{ kJ}$$



$$\begin{aligned} \Delta H_{\text{rxn}}^\circ &= \sum n \Delta H_{f,\text{products}}^\circ - \sum m \Delta H_{f,\text{reactants}}^\circ \\ &= 2 \text{ mol CO}_2 \times \left(\frac{-393.5 \text{ kJ}}{1 \text{ mol CO}_2} \right) - \left[2 \text{ mol CO} \times \frac{-110.5 \text{ kJ}}{1 \text{ mol CO}} + 1 \text{ mol O}_2 \times \frac{0 \text{ kJ}}{1 \text{ mol O}_2} \right] \end{aligned}$$

Hess's Law: Determine $\Delta H_{\text{rxn}}^\circ$ Indirectly

- $\Delta H_{\text{rxn}}^\circ = \sum \textcolor{red}{n} \Delta H_{f,\text{products}}^\circ - \sum \textcolor{red}{m} \Delta H_{f,\text{reactants}}^\circ$

Calculate the standard heat of reaction ($\Delta H_{\text{rxn}}^\circ$) for the following reaction:



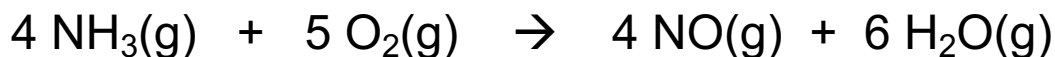
Reactant or product	ΔH_f° (kJ/mol, from Appendix IIB)
$\text{NH}_3(\text{g})$	-45.9
$\text{O}_2(\text{g})$	0.0
$\text{NO}(\text{g})$	+91.3
$\text{H}_2\text{O}(\text{g})$	-241.8

$$\begin{aligned} \Delta H_{\text{rxn}}^\circ &= \textcolor{red}{4 \text{mol NO} \times \left(\frac{+91.3 \text{ kJ}}{1 \text{mol NO}} \right)} + \textcolor{red}{6 \text{mol H}_2\text{O} \times \left(\frac{-241.8 \text{ kJ}}{1 \text{mol H}_2\text{O}} \right)} \\ &\quad - \textcolor{blue}{4 \text{mol NH}_3(\text{g}) \times \left(\frac{-45.9 \text{ kJ}}{1 \text{mol NH}_3} \right)} - \textcolor{blue}{5 \text{mol O}_2 \times \left(\frac{0.0 \text{ kJ}}{1 \text{mol O}_2} \right)} \\ &= \underline{\underline{-902.0 \text{ kJ}}} \\ &\quad \textcolor{blue}{1 \text{mol rxn}} \end{aligned}$$

Hess's Law: Determine $\Delta H_{\text{rxn}}^\circ$ Indirectly

- $\Delta H_{\text{rxn}}^\circ = \sum n \Delta H_{f,\text{products}}^\circ - \sum m \Delta H_{f,\text{reactants}}^\circ$

Calculate the standard heat of reaction ($\Delta H_{\text{rxn}}^\circ$) for the following reaction:



Reactant or product	ΔH_f° (kJ/mol, from Appendix IIB)
$\text{NH}_3(\text{g})$	-45.9
$\text{O}_2(\text{g})$	0.0
$\text{NO}(\text{g})$	+91.3
$\text{H}_2\text{O}(\text{g})$	-241.8

$$\begin{aligned}
 \Delta H_{\text{rxn}}^\circ &= 4 \text{ mol NO} \times \frac{91.3 \text{ kJ}}{1 \text{ mol NO}} + 6 \text{ mol H}_2\text{O} \times \frac{-241.8 \text{ kJ}}{1 \text{ mol H}_2\text{O}} - 4 \text{ mol NH}_3 \times \frac{-45.9 \text{ kJ}}{1 \text{ mol NH}_3} \\
 &\quad - 5 \text{ mol O}_2 \times \frac{0.0 \text{ kJ}}{1 \text{ mol O}_2} \\
 &= -902.0 \text{ kJ}
 \end{aligned}$$

Your Turn! $\Delta H_{rxn}^{\circ} = \sum n \Delta H_{f,products}^{\circ} - \sum m \Delta H_{f,reactants}^{\circ}$

Calculate the standard heat of reaction (ΔH_{rxn}°) for the following reaction:



$$\Delta H_{rxn}^{\circ} =$$

TABLE 9.4 Standard Enthalpies (K)

Formula	ΔH_f° (kJ/mol)	Formula	ΔH_f° (kJ/mol)
Bromine		Oxygen	
Br(g)	111.9	O ₂ (g)	0
Br ₂ (l)	0	O ₃ (g)	142.7
HBr(g)	-36.3	H ₂ O(g)	-241.8
Calcium		H ₂ O(l)	-285.8
Ca(s)	0	Silver	
CaO(s)	-634.9	Ag(s)	0
CaCO ₃ (s)	-1207.6	AgCl(s)	-127.0
Carbon		Sodium	
C(s, graphite)	0	Na(s)	0
C(s, diamond)	1.88	Na(g)	107.5
CO(g)	-110.5	NaCl(s)	-411.2
CO ₂ (g)	-393.5	Na ₂ CO ₃ (s)	-1130.7
CH ₄ (g)	-74.6	NaHCO ₃ (s)	-950.8
CH ₃ OH(l)	-238.6	Sulfur	
C ₂ H ₂ (g)	227.4	S ₈ (s, rhombic)	0

-2600.4 kJ

Your Turn! $\Delta H_{rxn}^{\circ} = \sum n \Delta H_{f,products}^{\circ} - \sum m \Delta H_{f,reactants}^{\circ}$

Calculate the standard heat of reaction (ΔH_{rxn}°) for the following reaction:



$$\Delta H_{rxn}^{\circ} = 4\text{mol CO}_2 \times \frac{-393.5\text{kJ}}{1\text{mol CO}_2} + 2\text{mol H}_2\text{O} \times \frac{-285.8\text{kJ}}{1\text{mol H}_2\text{O}} - 2\text{mol C}_2\text{H}_2 \times \frac{+227.4\text{kJ}}{1\text{mol C}_2\text{H}_2} - 5\text{mol O}_2 \times \frac{0\text{kJ}}{1\text{mol O}_2}$$

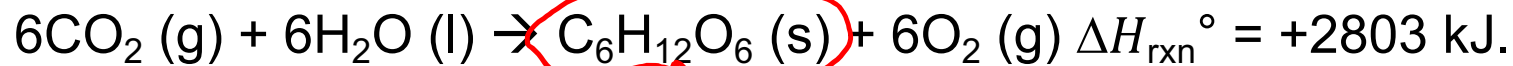
$$= -2600.4\text{kJ}$$

-2600.4 kJ

TABLE 9.4 Standard Enthalpies (K)

Formula	ΔH_f° (kJ/mol)	Formula	ΔH_f° (kJ/mol)
Bromine		Oxygen	
Br(g)	111.9	O ₂ (g)	0
Br ₂ (l)	0	O ₃ (g)	142.7
HBr(g)	-36.3	H ₂ O(g)	-241.8
Calcium		H₂O(l)	
Ca(s)	0		-285.8
Silver		Silver	
CaO(s)	-634.9	Ag(s)	0
CaCO ₃ (s)	-1207.6	AgCl(s)	-127.0
Carbon		Sodium	
C(s, graphite)	0	Na(s)	0
C(s, diamond)	1.88	Na(g)	107.5
CO(g)	-110.5	NaCl(s)	-411.2
CO ₂ (g)	-393.5	Na ₂ CO ₃ (s)	-1130.7
CH ₄ (g)	-74.6	NaHCO ₃ (s)	-950.8
CH ₃ OH(l)	-238.6	Sulfur	
C ₂ H ₂ (g)	227.4	S ₈ (s, rhombic)	0

Your Turn! $\Delta H_{rxn}^{\circ} = \sum n\Delta H_{f,products}^{\circ} - \sum m\Delta H_{f,reactants}^{\circ}$



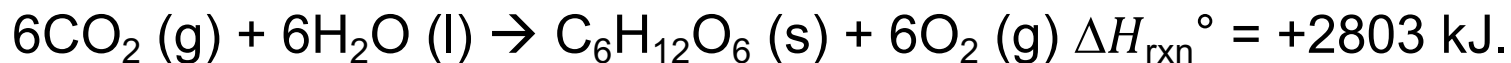
Determine the heat of formation (ΔH_f°) for glucose and write the equation.

TABLE 9.4 Standard Enthalpies (K)

Formula	$\Delta H_f^{\circ}(\text{kJ/mol})$	Formula	$\Delta H_f^{\circ}(\text{kJ/mol})$
Bromine		Oxygen	
Br(g)	111.9	O ₂ (g)	0
Br ₂ (l)	0	O ₃ (g)	142.7
HBr(g)	-36.3	H ₂ O(g)	-241.8
Calcium		Silver	
Ca(s)	0	H ₂ O(l)	-285.8
CaO(s)	-634.9	Ag(s)	0
CaCO ₃ (s)	-1207.6	AgCl(s)	-127.0
Carbon		Sodium	
C(s, graphite)	0	Na(s)	0
C(s, diamond)	1.88	Na(g)	107.5
CO(g)	-110.5	NaCl(s)	-411.2
CO ₂ (g)	-393.5	Na ₂ CO ₃ (s)	-1130.7
CH ₄ (g)	-74.6	NaHCO ₃ (s)	-950.8
CH ₃ OH(l)	-238.6	Sulfur	
C ₂ H ₂ (g)	227.4	S ₈ (s, rhombic)	0

-1273 kJ/mol

Your Turn! $\Delta H_{rxn}^{\circ} = \sum n\Delta H_{f,products}^{\circ} - \sum m\Delta H_{f,reactants}^{\circ}$



Determine the heat of formation (ΔH_f°) for glucose and write the equation.

$$1 \text{ mol C}_6\text{H}_{12}\text{O}_6 \times \Delta H_f^{\circ} + 6 \text{ mol O}_2 \times \frac{0 \text{ kJ}}{1 \text{ mol O}_2} - 6 \text{ mol H}_2\text{O} \times \frac{-285.8 \text{ kJ}}{1 \text{ mol H}_2\text{O}} - 6 \text{ mol CO}_2 \times \frac{-393.5 \text{ kJ}}{1 \text{ mol CO}_2} = +2803 \text{ kJ}$$

$$\Delta H_f^{\circ} = -1273 \frac{\text{kJ}}{\text{mol}}$$

TABLE 9.4 Standard Enthalpies (K)

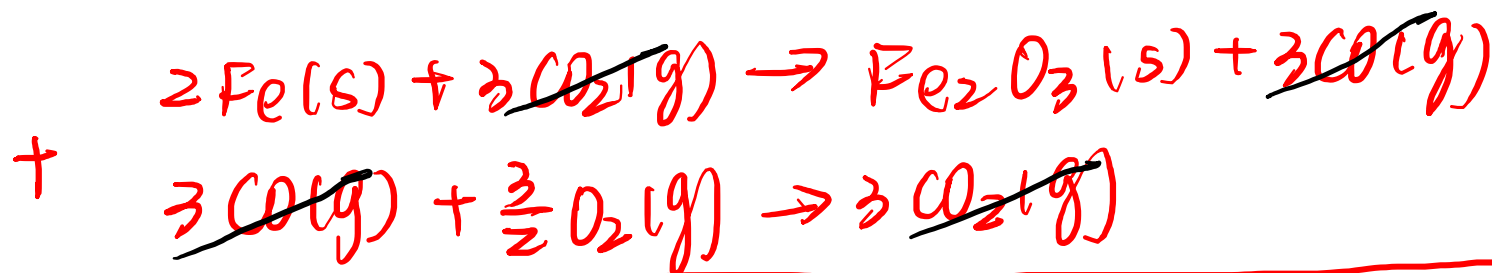
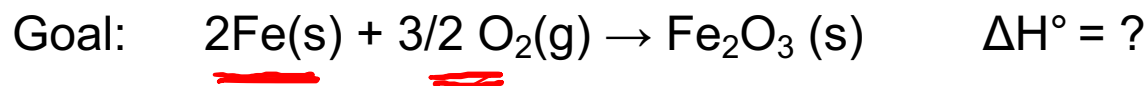
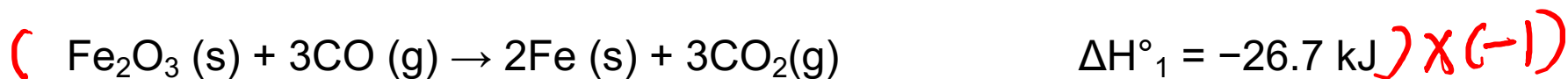
Formula	ΔH_f° (kJ/mol)	Formula	ΔH_f° (kJ/mol)
Bromine		Oxygen	
Br(g)	111.9	O ₂ (g)	0
Br ₂ (l)	0	O ₃ (g)	142.7
HBr(g)	-36.3	H ₂ O(g)	-241.8
Calcium		H₂O(l)	
Ca(s)	0		-285.8
CaO(s)	-634.9	Silver	
CaCO ₃ (s)	-1207.6	Ag(s)	0
Carbon		AgCl(s)	-127.0
C(s, graphite)	0	Sodium	
C(s, diamond)	1.88	Na(s)	0
CO(g)	-110.5	Na(g)	107.5
CO ₂ (g)	-393.5	NaCl(s)	-411.2
CH ₄ (g)	-74.6	Na ₂ CO ₃ (s)	-1130.7
CH ₃ OH(l)	-238.6	NaHCO ₃ (s)	-950.8
Sulfur		Sulfur	
C ₂ H ₂ (g)	227.4	S ₈ (s, rhombic)	0

-1273 kJ/mol

Hess's Law: Determine $\Delta H_{\text{rxn}}^\circ$ Indirectly

Calculate the standard heat of reaction ($\Delta H_{\text{rxn}}^\circ$) from related chemical equations.

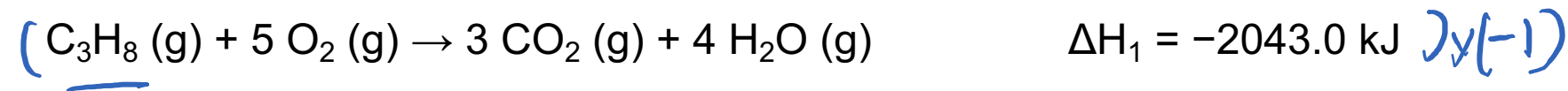
Knowing:



$$\Delta H_{\text{rxn}}^\circ = (-26.7 \text{ kJ}) \times (-1) + (-283.0 \text{ kJ}) \times 3 = -822.3 \text{ kJ}$$

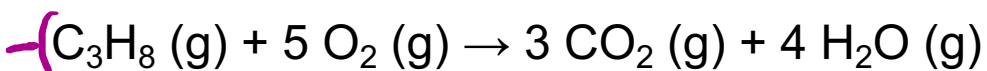
Your Turn!

Find the ΔH for the reaction: 3C (s) + 4 H₂ (g) $\rightarrow$ C₃H₈ (g) given the following thermochemical equations:



Your Turn!

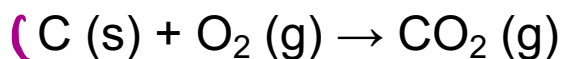
Find the ΔH for the reaction: $3\text{C (s)} + 4\text{H}_2\text{(g)} \rightarrow \text{C}_3\text{H}_8\text{(g)}$ given the following thermochemical equations:



$$\Delta H_1 = -2043.0 \text{ kJ}$$



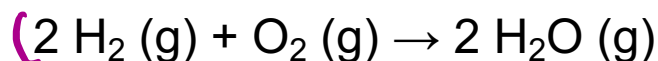
$$\Delta H'_1 = +2043.0 \text{ kJ}$$



$$\Delta H_2 = -393.5 \text{ kJ} \times 3$$



$$\Delta H'_2 = -1180.5 \text{ kJ}$$



$$\Delta H_3 = -483.6 \text{ kJ} \times 2$$

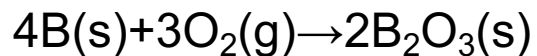


$$\Delta H'_3 = -967.2 \text{ kJ}$$

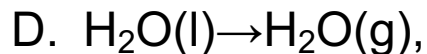
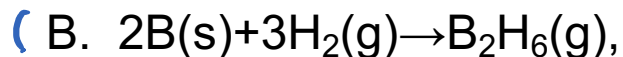
$$\Delta H_{\text{rxn}} = +2043.0 \text{ kJ} - 1180.5 \text{ kJ} - 967.2 \text{ kJ} = -104.7 \text{ kJ}$$

Your Turn!

Calculate the enthalpy of the reaction:



given the following pertinent information:



$$\Delta H_A = +2035 \text{ kJ }) \times (-2)$$

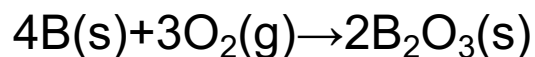
$$\Delta H_B = +36 \text{ kJ }) \times 2$$

$$\Delta H_C = -285 \text{ kJ}$$

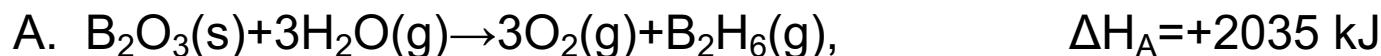
$$\Delta H_D = +44 \text{ kJ}$$

Your Turn!

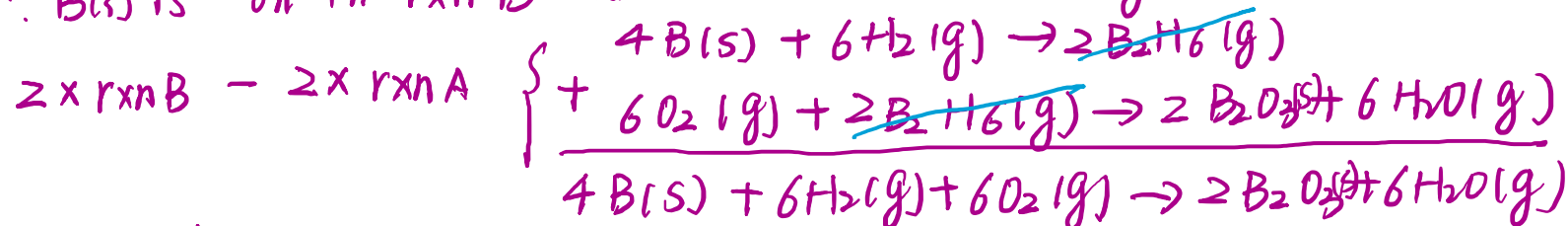
Calculate the enthalpy of the reaction:



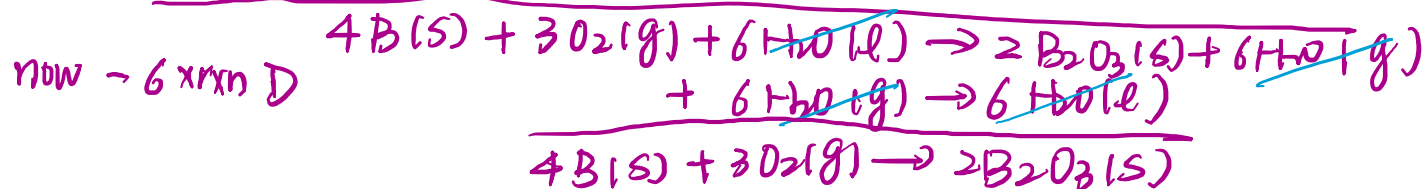
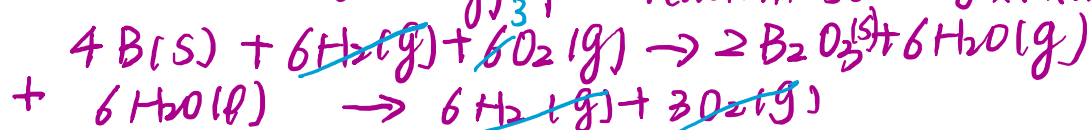
given the following pertinent information:



$\therefore \text{B(s)}$ is on in rxn B and $\text{B}_2\text{O}_3\text{(s)}$ is only in rxn A



it needs to remove $6\text{H}_2\text{(g)}$ from reactant so $-6 \times \text{rxn C}$



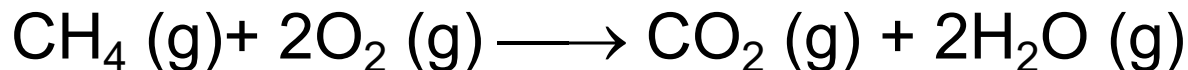
$$\Delta H = 2 \times (+36 \text{ kJ}) - 2 \times (+2035 \text{ kJ}) - 6 \times (-285 \text{ kJ}) - 6 \times (+44 \text{ kJ}) = -2552 \text{ kJ}$$

-2552.7 kJ

Estimating ΔH_{rxn} from Bond Energies

Bond Energy: the amount of energy required to break one mole of a bond in a compound in the **gas state**.

- **Break** a bond (Reactants): Absorb / release energy therefore $\Delta H_{\text{bonds}} _ 0$ ()
- **Form** a bond (Products): Absorb / release energy therefore $\Delta H_{\text{bonds}} _ 0$ ()



In one mol rxn:	Before reaction (Reactants)	After reaction (Products)
Chemical Bonds	_____ mol of _____ _____ mol of _____	_____ mol of _____ _____ mol of _____

$$\Delta H_{\text{rxn}} = [+ \sum (\Delta H_{\text{bonds broken}})] + [- \sum (\Delta H_{\text{bonds formed}})]$$

Bond	BE (kJ/mol)
C—H	414
O=O	498
C=O	799
O—H	464

- Endothermic / exothermic?

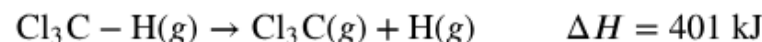
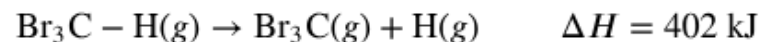
Table of Average Bond Energies

TABLE 9.3 Average Bond Energies

Bond	Bond Energy (kJ/mol)	Bond	Bond Energy (kJ/mol)	Bond	Bond Energy (kJ/mol)
H—H	436	N—N	163	Br—F	237
H—C	414	N=N	418	Br—Cl	218
H—N	389	N≡N	946	Br—Br	193
H—O	464	N—O	222	I—Cl	208
H—S	368	N=O	590	I—Br	175
H—F	565	N—F	272	I—I	151
H—Cl	431	N—Cl	200	Si—H	323
H—Br	364	N—Br	243	Si—Si	226
H—I	297	N—I	159	Si—C	301
C—C	347	O—O	142	S—O	265
C=C	611	O=O	498	Si=O	368
C≡C	837	O—F	190	S=O	523
C—N	305	O—Cl	203	Si—Cl	464
C=N	615	O—I	234	S=S	418
C≡N	891	F—F	159	S—F	327
C—O	360	Cl—F	253	S—Cl	253
C=O	736*	Cl—Cl	243	S—Br	218
C≡O	1072			S—S	266
C—Cl	339				

*799 in CO₂

- Bond energies are measured in experiments.
- Bond energy for a bond is _____ out through many chemical compounds that contain that bond.



- Higher bond order, _____ bond energy and _____ bonds
 $\text{C}\equiv\text{C}$ (837 kJ)
 $\text{C}=\text{C}$ (611 kJ)
 $\text{C}-\text{C}$ (347 kJ)
- The double bond is not directly two times of the single bond.
 (Typically, π is _____ than σ)

Your Turn! $\Delta H_{\text{rxn}} = [+ \sum (\Delta H_{\text{bonds broken}})] + [- \sum (\Delta H_{\text{bonds formed}})]$

Using bond energies, calculate ΔH_{rxn} for : $\text{CH}_4(\text{g}) + 2 \text{H}_2\text{O}(\text{g}) \rightarrow 4 \text{H}_2(\text{g}) + \text{CO}_2(\text{g})$

Hint: You will need to write the Lewis structures to determine the type of bonds.

$\Delta H_{\text{rxn}} =$

TABLE 9.3 Average Bond Energies

Bond	Bond Energy (kJ/mol)	Bond	Bond Energy (kJ/mol)
H—H	436	N—N	163
H—C	414	N=N	418
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H—Cl	431	N—Cl	200
H—Br	364	N—Br	243
H—I	297	N—I	159
C—C	347	O—O	142
C=C	611	O=O	498
C≡C	837	O—F	190
C—N	305	O—Cl	203
C=N	615	O—I	234
C≡N	891	F—F	159
C—O	360	Cl—F	253
C=O	736*	Cl—Cl	243
C≡O	1072		
C—Cl	339		

*799 in CO_2

Your Turn! $\Delta H_{\text{rxn}} = [+ \sum (\Delta H_{\text{bonds broken}})] + [- \sum (\Delta H_{\text{bonds formed}})]$

Using bond energies, calculate ΔH_{rxn} for : $\text{COCl}_2(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{HCl}(\text{g})$

Hint: You will need to write the Lewis structures to determine the type of bonds.

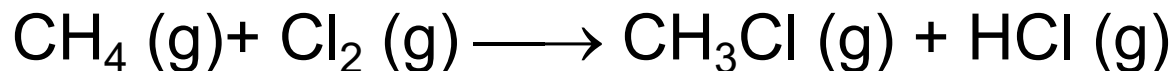
$\Delta H_{\text{rxn}} =$

TABLE 9.3 Average Bond Energies

Bond	Bond Energy (kJ/mol)	Bond	Bond Energy (kJ/mol)
H—H	436	N—N	163
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*799 in CO_2

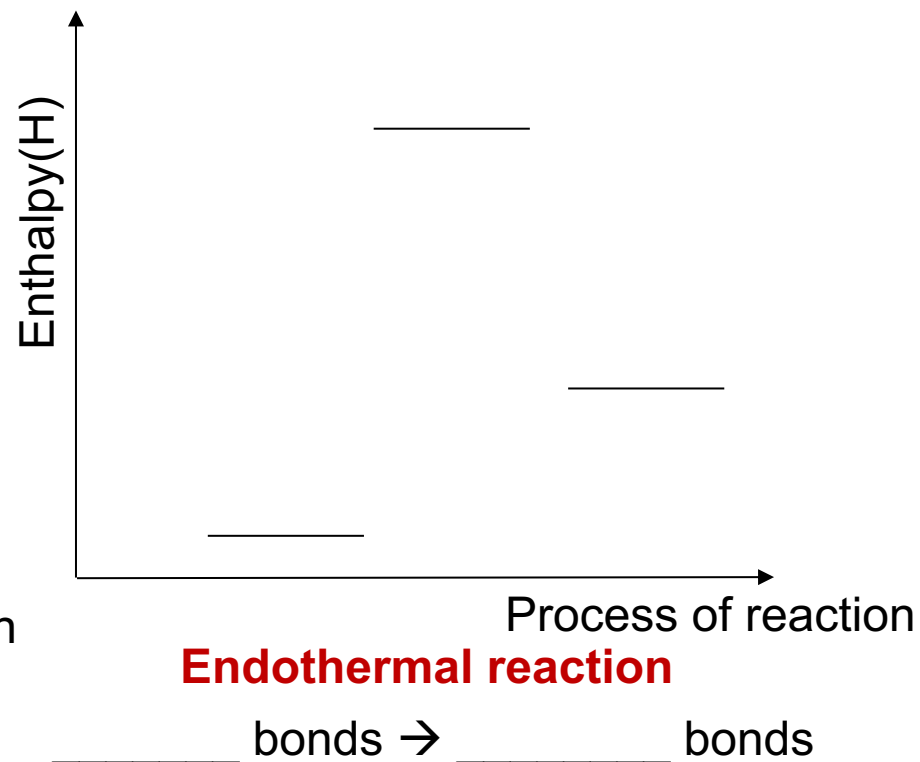
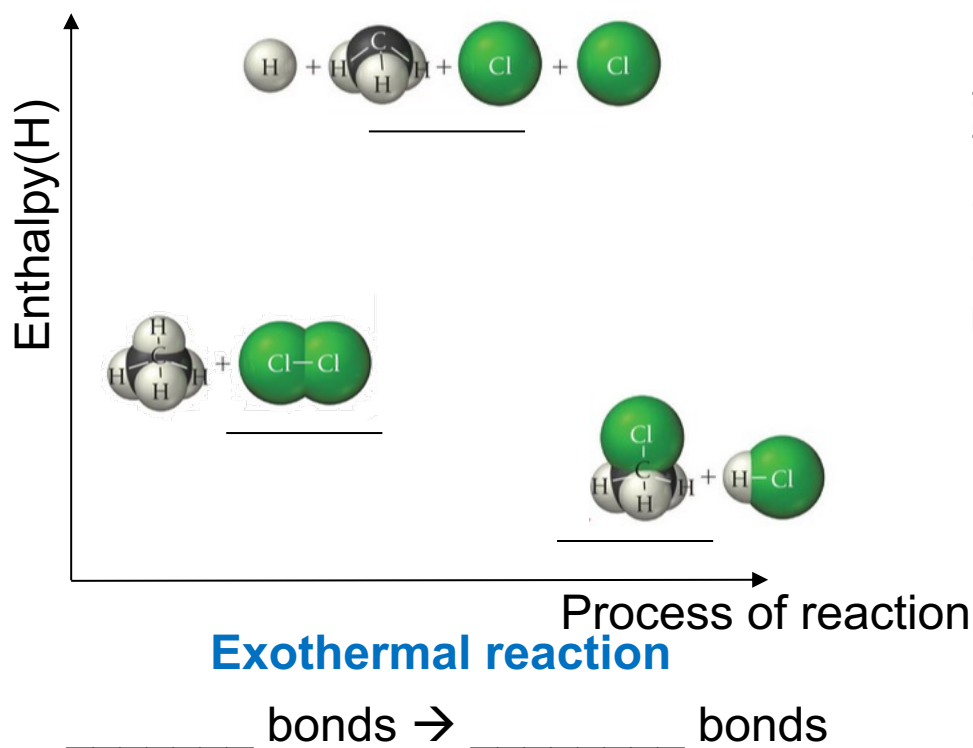
Estimating ΔH_{rxn} from Bond Energies



Bond	BE (kJ/mol)
C—H	414
Cl—Cl	243
C—Cl	339
Cl—H	431

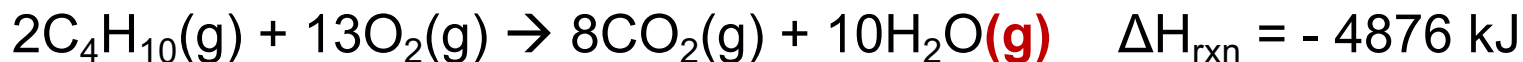
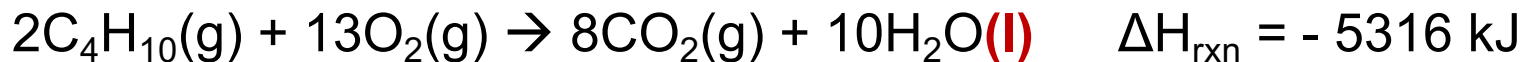
$$\Delta H_{\text{rxn}} = \sum (\Delta H_{\text{bonds broken}}) + \sum (\Delta H_{\text{bonds formed}})$$

Reaction coordinate diagram:



Your Turn!

Use the information below:



Which physical state of water is more stable?

- A. $\text{H}_2\text{O}(\text{l})$
- B. They are the same
- C. $\text{H}_2\text{O}(\text{g})$

